

Machine Learning Models Summary

Category	Model Type	Examples	Usage
Linear Models	Regression	Linear Regression, Ridge, Lasso, ElasticNet, SGDRegressor	Predict continuous values
Linear Models	Classification	Logistic Regression, LinearSVC, SGDClassifier	Binary/multi-class prediction
Linear Models	Bayesian	BayesianRidge, ARDRegression	Probabilistic regression
Linear Models	Dimensionality Reduction	LDA	Lower dimensions + classification
Tree Models	Single Tree	DecisionTreeClassifier/Regressor	Interpretable tree models
Tree Models	Bagging	Random Forest, Extra Trees	Reduces variance, robust
Tree Models	Boosting	AdaBoost, XGBoost, LightGBM, CatBoost	High accuracy
Deep Learning	ANN	MLP, Dense Network	General deep learning
Deep Learning	CNN	ResNet, VGG, EfficientNet	Images, vision tasks
Deep Learning	RNN	LSTM, GRU	Text, time series
Deep Learning	Transformers	BERT, GPT, T5	Modern NLP models
Clustering	Partitioning	K-Means, K-Medoids	Grouping large datasets
Clustering	Density-based	DBSCAN, HDBSCAN	Irregular shapes, noise
Clustering	Model-based	Gaussian Mixture Model	Probabilistic clusters